KAIZEN MATHS

Lesson starter

Integration of Algebraic Fractions

Quadratic Denominators

Name _____

S1

Recognise when the numerator is the derivative of the quadratic denominator.

Find $\int \frac{8x-12}{x^2-3x+6}\,dx$.

S2

Use the standard inverse tangent integral form.

Find $\int \frac{2}{x^2+4}\,dx$.

S3

Complete the square, then use the inverse tangent form.

Find $\int \frac{3}{x^2+8x+52}\,dx$.

S4

Split the numerator into a derivative part and a constant part.

Find $\int \frac{2x+9}{x^2+2x+5}\,dx$.

Show your working

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ALG-31E3-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Integration of Algebraic Fractions

Quadratic Denominators

Name

E1

Recognise when the numerator is the derivative of the quadratic denominator.

Find $\int \frac{6x+18}{x^2+6x+7}\,dx$.

E2

Use the standard inverse tangent integral form.

Find $\int \frac{8}{x^2+36}\,dx$.

E3

Complete the square, then use the inverse tangent form.

Find $\int \frac{8}{x^2+6x+13}\,dx$.

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ALG-31E3-L4 · E

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Integration of Algebraic Fractions

Quadratic Denominators

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Recognise when the numerator is the derivative of the quadratic denominator.

Find $\int \frac{6x+18}{x^2+6x+7}\,dx$.

E2

Use the standard inverse tangent integral form.

Find $\int \frac{8}{x^2+36}\,dx$.

E3

Complete the square, then use the inverse tangent form.

Find $\int \frac{8}{x^2+6x+13}\,dx$.

ALG-31E3-L4 · E

Show your working

Teacher answers & working

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S1

Recognise when the numerator is the derivative of the quadratic denominator.

Find $\int \frac{8x-12}{x^2-3x+6}\,dx$.

Answer:

$4\ln|x^2 - 3x + 6| + C$

Working:

Let $f(x) = x^2 - 3x + 6$.

Then f'(x) = 2x − 3.

The numerator is 4f'(x).

$\int \frac{4f'(x)}{f(x)}\,dx = 4\ln|f(x)| + C$.

Therefore the answer is $4\ln|x^2 - 3x + 6| + C$.

S2

Use the standard inverse tangent integral form.

Find $\int \frac{2}{x^2+4}\,dx$.

Answer:

$1\tan^{-1}\frac{x}{2} + C$

Working:

Use the standard result $\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$.

Here $a^2 = 4$, so a = 2.

$\int \frac{2}{x^2+4}\,dx = 2\int \frac{1}{x^2+4}\,dx$.

$= 1\tan^{-1}\frac{x}{2} + C$.

S3

Complete the square, then use the inverse tangent form.

Find $\int \frac{3}{x^2+8x+52}\,dx$.

Answer:

$\frac{1}{2}\tan^{-1}\frac{x+4}{6} + C$

Working:

Complete the square in the denominator.

$x^2 + 8x + 52 = (x + 4)^2 + 36$.

Use $u = x + 4$, so $du = dx$.

The integral becomes $\int \frac{3}{u^2+36}\,du$.

$= \frac{1}{2}\tan^{-1}\frac{u}{6} + C$.

Substitute back u = x + 4: $\frac{1}{2}\tan^{-1}\frac{x+4}{6} + C$.

Teacher answers & working

Integration of Algebraic Fractions · Quadratic Denominators

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S4

Split the numerator into a derivative part and a constant part.

Find $\int \frac{2x+9}{x^2+2x+5}\,dx$.

Answer:

$1\ln(x^2 + 2x + 5) + \frac{7}{2}\tan^{-1}\frac{x+1}{2} + C$

Working:

Complete the square: $x^2 + 2x + 5 = (x + 1)^2 + 4$.

The derivative of the denominator is $2(x + 1) = 2x + 2$.

Rewrite the numerator as $1[2x + 2] + 7$.

Split the integral into two parts.

$\int \frac{1(2x+2)}{x^2+2x+5}\,dx + \int \frac{7}{x^2+2x+5}\,dx$.

The first part is $1\ln(x^2 + 2x + 5)$.

The second part is $\frac{7}{2}\tan^{-1}\frac{x+1}{2}$.

Final answer: $1\ln(x^2 + 2x + 5) + \frac{7}{2}\tan^{-1}\frac{x+1}{2} + C$.

E1

Recognise when the numerator is the derivative of the quadratic denominator.

Find $\int \frac{6x+18}{x^2+6x+7}\,dx$.

Answer:

$3\ln|x^2 + 6x + 7| + C$

Working:

Let $f(x) = x^2 + 6x + 7$.

Then $f'(x) = 2x + 6$.

The numerator is $3f'(x)$.

$\int \frac{3f'(x)}{f(x)}\,dx = 3\ln|f(x)| + C$.

Therefore the answer is $3\ln|x^2 + 6x + 7| + C$.

Teacher answers & working

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E2

Use the standard inverse tangent integral form.

Find $\int \frac{8}{x^2+36}\,dx$.

Answer:

$\frac{4}{3}\tan^{-1}\frac{x}{6} + C$

Working:

Use the standard result $\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$.

Here $a^2 = 36$, so $a = 6$.

$\int \frac{8}{x^2+36}\,dx = 8\int \frac{1}{x^2+36}\,dx$.

$= \frac{4}{3}\tan^{-1}\frac{x}{6} + C$.

E3

Complete the square, then use the inverse tangent form.

Find $\int \frac{8}{x^2+6x+13}\,dx$.

Answer:

$4\tan^{-1}\frac{x+3}{2} + C$

Working:

Complete the square in the denominator.

$x^2 + 6x + 13 = (x + 3)^2 + 4$.

Use $u = x + 3$, so $du = dx$.

The integral becomes $\int \frac{8}{u^2+4}\,du$.

$= 4\tan^{-1}\frac{u}{2} + C$.

Substitute back $u = x + 3$: $4\tan^{-1}\frac{x+3}{2} + C$.